Walmart Sales Analysis: A SQL-Driven Exploration

Optimizing Retail Strategy through Data Science — An exhaustive analysis of 1,000+ transactions across three branches using advanced SQL analytics.

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Executive Summary

This project analyzes a **1,000-row Walmart sales dataset** spanning three branches across Myanmar — Yangon, Naypyitaw, and Mandalay. Using structured SQL queries, we uncover revenue trends, product performance, customer behavior, and operational benchmarks to inform strategic retail decisions.

1,000+

Transactions analyzed

3 Branches

Across 3 cities

17 Columns

Rich data dimensions

6 Product Lines

Category-level insights

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The Data Ecosystem

The dataset comprises **17 structured columns** enabling multi-dimensional analysis across transactions, products, customers, and financials.

Identifiers

- Invoice ID
- Branch
- City

Product

- Product Line
- Unit Price
- Quantity

Customer

- Customer Type
- Gender
- Payment
- Rating

Time

- Date
- Time

Financials

- Tax (5%)
- Total
- COGS
- Gross Income
- Gross Margin %

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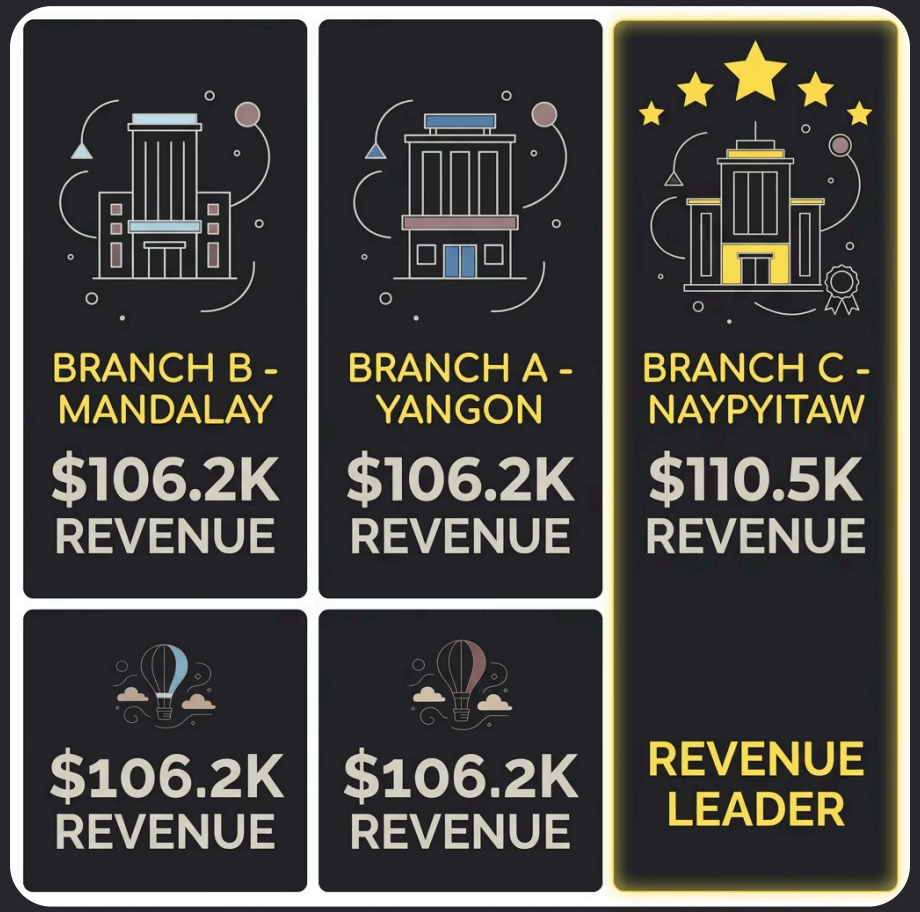
Geographic Distribution



Branch Revenue Snapshot

Branch C — Naypyitaw leads with **$110,500** in total revenue, edging out Yangon and Mandalay, which each generated approximately **$106,200**.

While geographic spread is balanced, Naypyitaw's performance signals stronger average transaction values and customer volume — a strategic focus area for inventory and staffing investment.

Branch C (Naypyitaw) is the clear revenue leader across all three markets.

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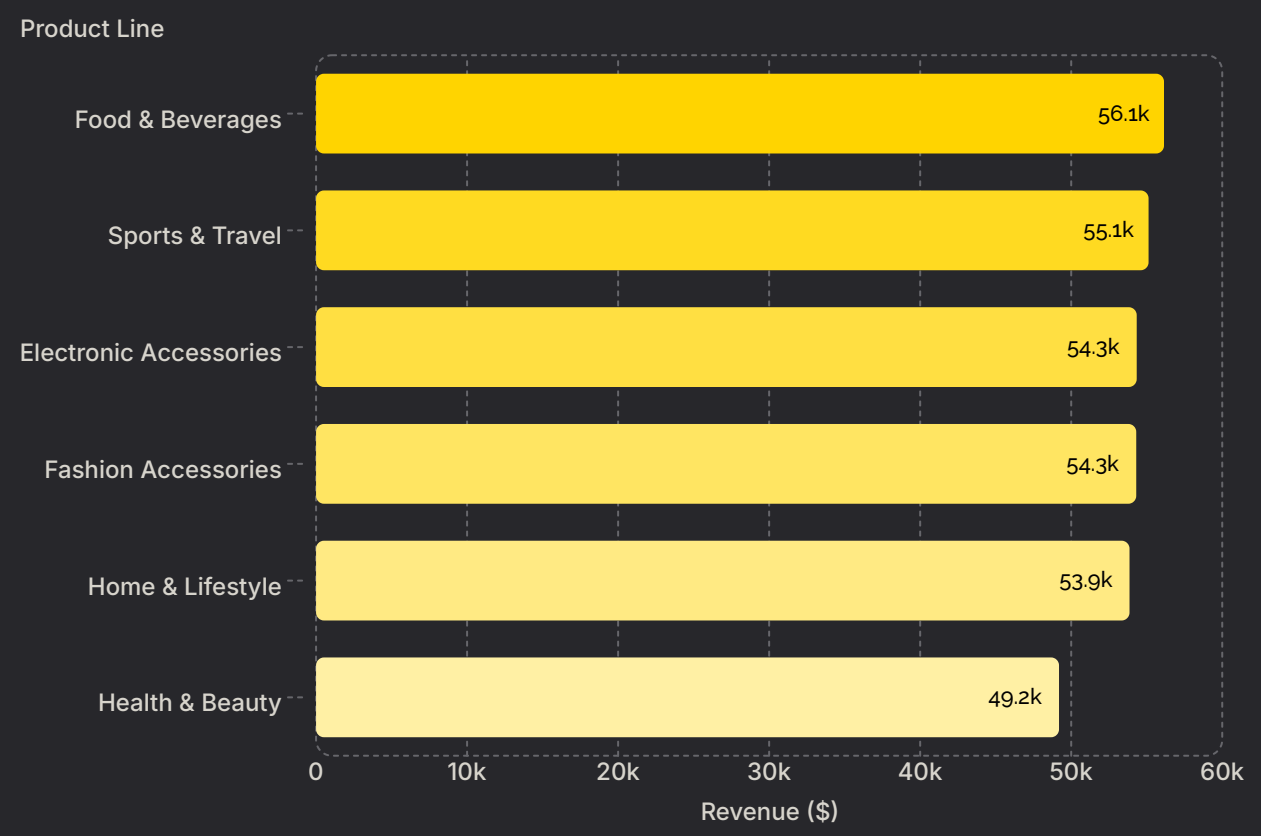
Product Line Performance



Top Categories Drive Revenue

Food & Beverages leads at **$56K**, closely followed by **Sports & Travel** at **$55K**. The competitive spread across most lines suggests balanced demand.

Health & Beauty lags behind — presenting a merchandising and promotional opportunity to close the $7K revenue gap.

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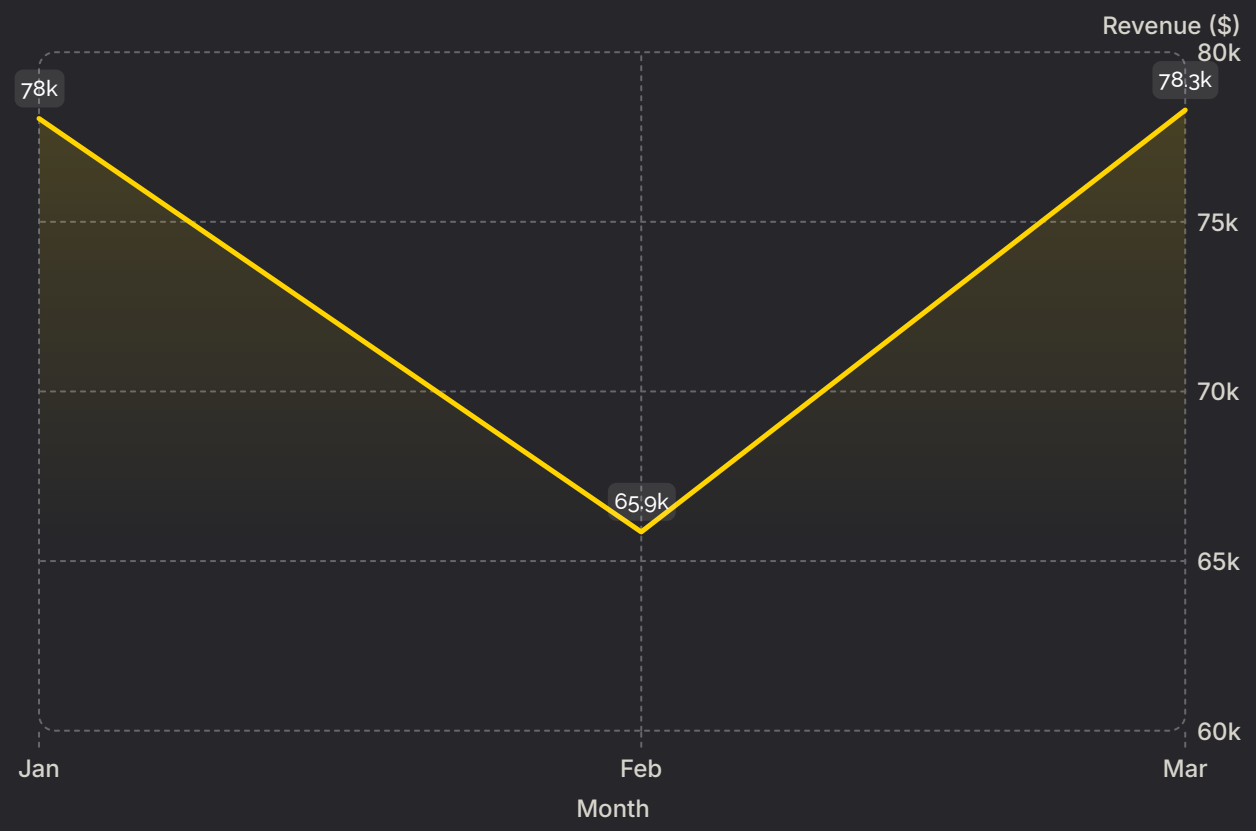
Monthly Trend Analysis



The February Dip Explained

Revenue rebounded strongly in March to **$78,295** after a notable **14% decline** in February to $65,859.

Two structural factors drive this dip: February's **28-day calendar cycle** reduces total transaction days, while **post-holiday consumer cooldown** suppresses discretionary spending. This is a predictable, seasonal pattern — not a demand problem.

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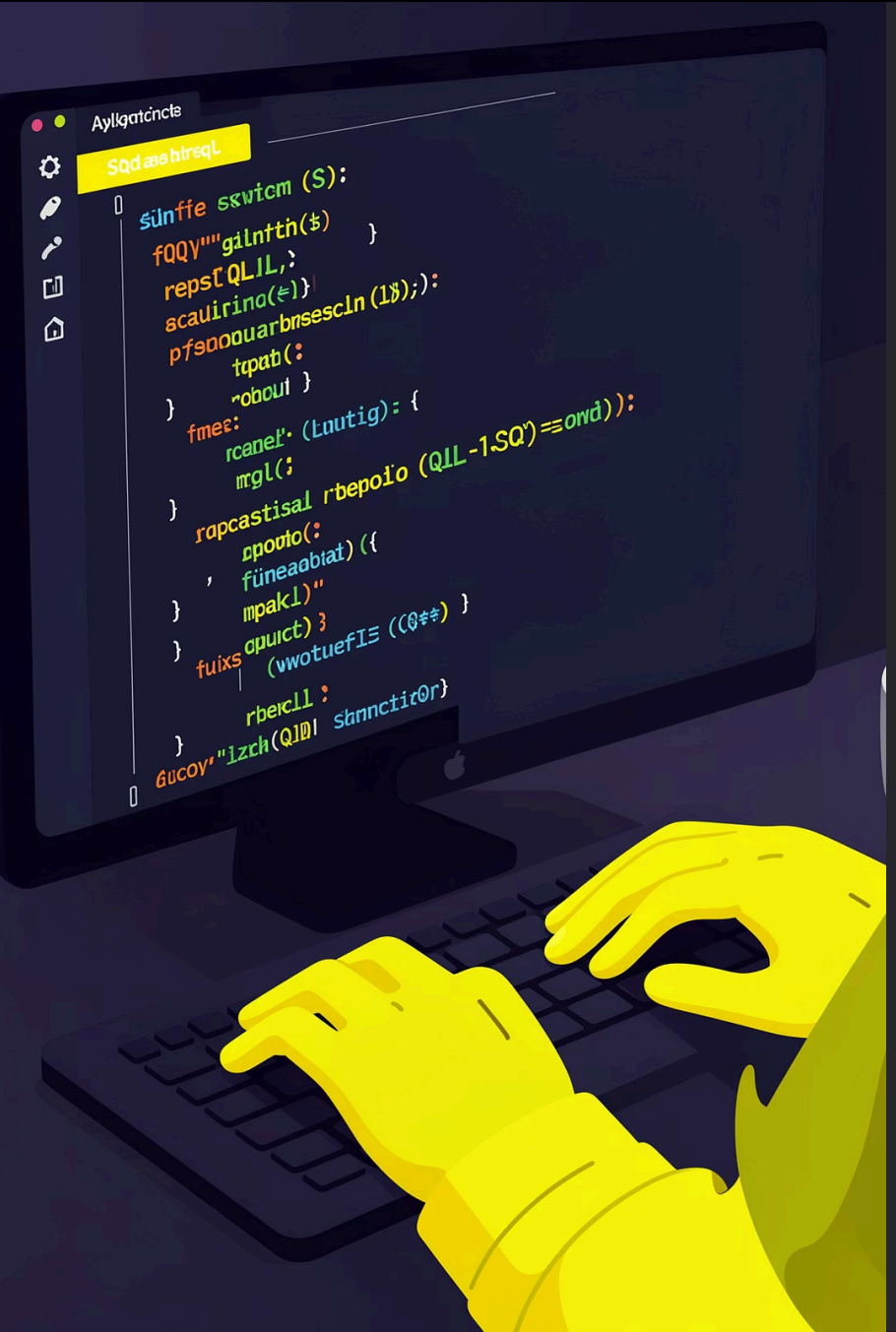

SQL Technical Spotlight

Performance Remarking Query — Classifying Branch Output

```sql
SELECT
    branch,
    AVG(quantity) AS avg_quantity,
    CASE
        WHEN AVG(quantity) > 5 THEN 'GOOD'
        ELSE 'BAD'
    END AS performance_flag
FROM walmart_sales
GROUP BY branch
ORDER BY avg_quantity DESC;
```

This query uses a `CASE` expression to dynamically classify each branch as **GOOD** or **BAD** based on average unit quantity sold — enabling rapid performance triage without manual calculation.

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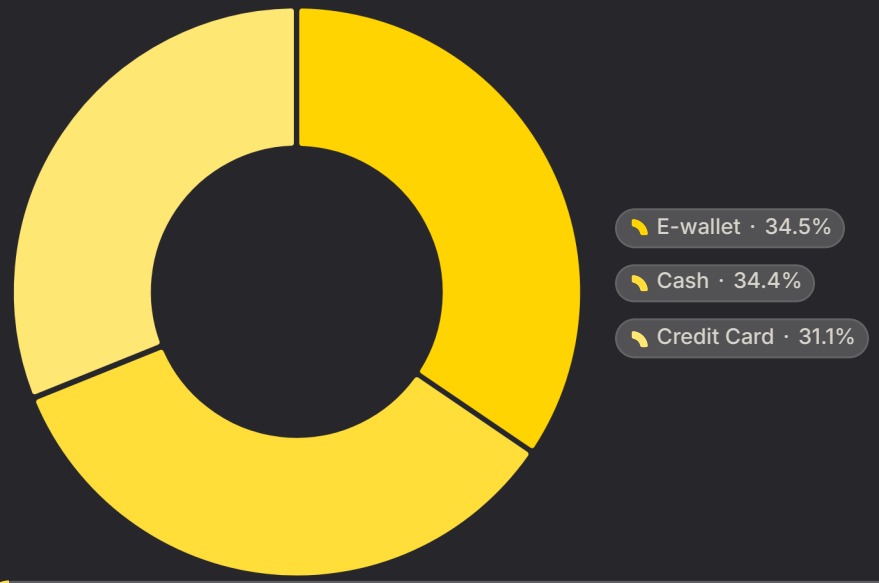
Customer Insights



Payment & Satisfaction Breakdown

Payment preferences are nearly **evenly distributed** across all three methods, indicating Walmart's customer base is diverse in financial behavior. Supporting all payment types is non-negotiable.

E-wallet — 34.5%

Fastest-growing digital payment segment

Food & Bev Rating: 7.11 ★

Highest-rated product category across all branches

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Advanced Benchmarking

Branch Efficiency Query — Filtering Above-Average Volume

```sql
SELECT
    branch,
    SUM(quantity)    AS total_units_sold,
    AVG(quantity)    AS avg_units_per_txn,
    COUNT(*)         AS total_transactions
FROM walmart_sales
GROUP BY branch
HAVING AVG(quantity) > (
    SELECT AVG(quantity)
    FROM walmart_sales
)
ORDER BY total_units_sold DESC;
```

The `HAVING` clause filters to branches whose per-transaction average **exceeds the global mean** — isolating high-efficiency locations for resource prioritization and performance benchmarking.

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Conclusion & Strategic Recommendations

🛒 Food & Beverages

Maintain high inventory levels year-round. This is the top revenue and top-rated category — no room for stockouts.

📅 February Strategy

Launch targeted promotions in Sports & Travel during February to offset the predictable post-holiday trough and short-month cycle.

📍 Branch C Focus

Invest in Naypyitaw's operational capacity — it's the revenue leader and should be the model for branch expansion planning.

ⓘ Key Takeaway: Data-driven retail strategy — not intuition — is what separates good decisions from great ones.

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